

# PORTFOLIO DECISION SYSTEM (PDS)

## Architecture, Governance, and Operating Framework

A framework for combining independently generated strategy targets into a single monitored portfolio

### SYSTEM DOCUMENTATION v1.5

This document describes the current three-layer PDS architecture: PDS Core, Reinforcement-Learning Adaptive Risk Control, and Dynamic FX. It also sets out the rules for decision timing, historical records, accounting, monitoring, and public reporting. Appendix D records the PDS-specific Adaptive implementation, frozen-policy design, and evidence boundary.

### SlackQuant Research

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[research.slackquant.com/systems/pds/](https://research.slackquant.com/systems/pds/)

## Document Summary

| Field                        | Details                                                                                                                                                                                                                                                          |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Document                     | PDS System Documentation                                                                                                                                                                                                                                         |
| Version                      | 1.5                                                                                                                                                                                                                                                              |
| Release date                 | 2026-09-27                                                                                                                                                                                                                                                       |
| System type                  | Portfolio Decision & Operating System                                                                                                                                                                                                                            |
| Public scope                 | Public view of the current Core, Adaptive, Preview, FX, Performance, and Portfolio information. Environment-specific infrastructure, brokerage-account data, and protected implementation details are excluded.                                                  |
| Current architecture         | Layer 1: PDS Core (F2R 10% + ADAA 90% at the current monthly top-level target). Layer 2: PDS Adaptive (Reinforcement-Learning Adaptive Risk Control). Layer 3: Dynamic FX, the monthly USD/KRW implementation overlay.                                           |
| Decision and rebalance cycle | Monthly signals; rebalance at the first U.S. trading-day close common to both strategies after the signal month ends.                                                                                                                                            |
| Performance measurement      | Daily returns are the performance source of record. Monthly and annual figures are derived from the continuous daily series, using 252 trading days for annualization.                                                                                           |
| Disclosure policy            | Public and local dashboards present the same reader-facing operating information. The public version excludes local paths, runtimes, caches, credentials, repository controls, debug tooling, brokerage-account information, and protected controller artifacts. |

### How this document is organized

Sections 1-4 cover the overview, architecture, strategy inputs, and Core construction. Sections 5-9 cover timing, performance, portfolio status, Dynamic FX, and governance. Sections 10-12 cover dashboard monitoring, operations, and public disclosure. Sections 13-15 cover release validation, related methods, and limitations. Appendices A-D summarize decision status, public resources, the v1.5 specification, and the PDS-specific Adaptive technical implementation.

## OVERVIEW

# 1. Executive Overview

Portfolio Decision System (PDS) combines target portfolios from independently developed investment strategies into a single portfolio process. It leaves each strategy's investment logic unchanged and standardizes only the portfolio-level rules for integration, timing, accounting, monitoring, and disclosure. The underlying strategy outputs remain separately identifiable.

PDS has three operating layers. PDS Core combines Forecast-to-Rank Allocation (F2R) and ADAA under the current strategic allocation. PDS Adaptive starts from Core and adds Reinforcement-Learning Adaptive Risk Control, a bounded risk-control layer rather than a standalone alpha or security-selection model. Dynamic FX is a separate currency layer that applies the same USD/KRW rule, execution clock, and cost schedule to the relevant Core or Adaptive USD return path. Keeping the layers separate makes attribution and decision responsibility clear.



## A useful distinction

F2R and ADAA source targets, the Core Official Decision, Adaptive decisions, Dynamic FX decisions, Preview, Forward Shadow, standalone provider records, and reconstructed PDS histories do not carry the same status. Core defines the base portfolio; Adaptive changes only its own risk budget; Dynamic FX changes only the currency layer.

## What PDS does

- Combines strategy targets into one portfolio using a consistent set of portfolio rules.
- Applies the same rebalance, turnover, cost, and performance rules when the strategies are evaluated together.
- Keeps the current Official Decision separate from Preview and Forward Shadow outputs.
- Retains the last valid Official Decision when required data or timing checks do not align.
- Publishes Core, Adaptive, Preview, FX, Performance, and Portfolio information after the release checks pass.
- Runs PDS Core and PDS Adaptive in parallel so Core remains unchanged while Adaptive risk control is evaluated and operated separately.

## What PDS does not do

- It does not redesign the investment logic of F2R or ADAA.
- It does not treat Preview, Forward Shadow, or research-only FX outputs as approved portfolio decisions.
- An Official Decision records the PDS target; it is not a broker fill or an account-level trade confirmation.
- The public PDS dashboard presents the same reader-facing operating information as the local dashboard while excluding local paths, runtimes, caches, credentials, repository controls, and debug tooling.
- PDS Adaptive does not overwrite or backfill PDS Core history, and an Adaptive failure does not change the Core portfolio decision.
- RL does not select securities or replace Core. It adjusts the Adaptive risk budget within the approved bounds.

## ARCHITECTURE

## 2. System Architecture

PDS has three layers. Approved F2R and ADAA targets form Core. Adaptive applies a separately controlled reinforcement-learning risk-control layer to Core. Dynamic FX adds a monthly currency overlay to the Core and Adaptive USD paths. The layers share timing and accounting rules where needed, while their decisions and historical records remain separate.

| Stage | Step                  | What happens                                                                                                                                                                                                                          |
|-------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01    | Strategy targets      | Each strategy produces a target portfolio according to its own rules.                                                                                                                                                                 |
| 02    | Common format         | Targets are converted to a consistent asset-and-weight format so they can be combined without changing the underlying strategy rules.                                                                                                 |
| 03    | Input checks          | Strategy identity, date, timing, completeness, and weight consistency are checked before use.                                                                                                                                         |
| 04    | Strategy review       | Research evidence and practical portfolio value are reviewed before a strategy is approved for a portfolio role.                                                                                                                      |
| 05    | Portfolio combination | Approved targets are combined using the current PDS allocation policy.                                                                                                                                                                |
| 06    | Portfolio decision    | The monthly PDS Core target is set under the common decision and execution clock. PDS Adaptive may apply its own risk budget, while Dynamic FX makes a separate monthly currency decision without changing the underlying USD target. |
| 07    | Monitoring and review | Core, Adaptive, Dynamic FX, Current MTD, Preview, Forward Shadow (where applicable), and completed performance are monitored under defined clocks, with each view retaining its own status.                                           |

### What PDS standardizes

PDS never recreates a missing strategy output. It uses only source targets that can be traced to the underlying strategy and pass the required date, timing, and completeness checks. If a required input cannot be aligned, the update is not adopted and the last valid Official Decision remains in place.

#### Independent strategies, common portfolio rules

F2R and ADAA retain their own investment logic. PDS standardizes only the elements that must be common at portfolio level: target format, timing, portfolio combination, performance measurement, and monitoring.

## How the components fit together

| System       | Role in PDS                                                                                 | What remains separate                                                                                                                                                                  |
|--------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F2R          | Produces a machine-learning-based cross-asset target portfolio used by PDS.                 | F2R retains its forecasting, ranking, and target-construction rules.                                                                                                                   |
| ADAA         | Produces a multi-asset target portfolio used by PDS.                                        | ADAA retains its allocation rules and target construction.                                                                                                                             |
| PDS          | Combines the strategy targets and applies the common portfolio rules.                       | PDS does not recreate or override the underlying strategy models.                                                                                                                      |
| PDS Adaptive | Applies Reinforcement-Learning Adaptive Risk Control as a bounded layer on top of PDS Core. | It is not a standalone alpha engine or a full portfolio-construction model. The current state may be public, while exact controller parameters and training artifacts remain internal. |
| Dynamic FX   | Applies the monthly USD/KRW overlay to Core or Adaptive USD paths.                          | The currency layer is separate from the underlying portfolio decision; spot-based monitoring does not represent broker-level forward/NDF execution or realized hedge P&L.              |

## STRATEGY INPUTS

### 3. Strategy Inputs and Responsibilities

F2R and ADAA remain responsible for their own calculations. PDS takes their dated target portfolios, verifies the strategy identity, dates, timing, and weights, and combines them under PDS rules. A missing strategy output is never recreated inside PDS.

#### Checks before a target is used

- The correct strategy, target date, and relevant version are identified.
- Signal month, holding month, market-data cutoff, and rebalance timing line up.
- Weights are complete, properly scaled, and traceable to the strategy output.
- Strategy-level costs and PDS rebalancing costs are kept separate so the same cost is not counted twice.
- Missing, stale, or conflicting outputs are not replaced by estimates; the last valid Official Decision remains in place.

#### Standalone strategy history and PDS reconstructed history

F2R and ADAA retain their own standalone completed performance records: F2R from 1 May 2017 onward under its current model, and ADAA from 2 June 2008 onward under its standalone strategy record. PDS also maintains a separate retrospective reconstruction under the current PDS definition for cross-system comparison, applying the current provider definitions and PDS timing and accounting rules across the available historical period. This longer PDS series is a research and comparison record; it does not replace either standalone record or imply that the current PDS policy was operated historically. Adaptive fixed-policy comparison history is likewise kept separate from its operational decision ledger.

| Record                     | Purpose                                                                                                                                         | How it is used                                                                                                                                                 |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Standalone strategy record | Preserves each provider's own current-model or current-strategy completed performance record, including its native timing and cost conventions. | F2R standalone completed performance starts 1 May 2017; ADAA standalone performance starts 2 June 2008. These records are not replaced by PDS reconstructions. |

| Record                                | Purpose                                                                                                                                            | How it is used                                                                                                                                                                           |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PDS current-definition reconstruction | Applies the current provider definitions and PDS portfolio rules retrospectively over the available historical period for cross-system comparison. | The Core / Core + Dynamic FX common reader period is 2 December 2005 through 31 August 2026 for this release. It is research and comparison history, not a historical live-policy claim. |
| Public dashboard                      | Publishes the current reader-facing operating information after the release checks and removal of environment-specific details.                    | Used to show the current monitored state and completed performance without exposing environment-specific infrastructure.                                                                 |
| Adaptive operational history          | Post-deployment decisions and realized accounting only.                                                                                            | Separate from fixed-policy reconstructed comparison history and PDS Core; no retrospective splice or backfill.                                                                           |

### Historical and operational records are kept separate

Standalone F2R and ADAA records, the retrospective PDS current-definition reconstruction, Adaptive fixed-policy comparison history, Adaptive operational decisions, and the public operating view remain separately identifiable. Publication does not turn one series into another.

## CORE PORTFOLIO

### 4. Current Core Construction

PDS Core is the base portfolio. Its current monthly top-level target is F2R 10% and ADAA 90%. Each strategy determines its own asset weights; PDS combines the source targets under the common monthly execution, turnover, cost, and daily-accounting rules. The SlackQuant landing page emphasizes the architecture rather than the exact ratio; the dashboard and this document provide the construction detail.

PDS Core monthly target = F2R 10% + ADAA 90%

Total target weight = 100%

The ratio is part of the detailed system specification. The SlackQuant landing page emphasizes the architecture rather than the fixed mix.

The provider allocation is set at each monthly rebalance, not reset every day. After rebalancing, both provider weights and underlying asset weights drift with market returns until the next scheduled rebalance.

### PDS Adaptive

PDS Adaptive is a parallel Core-based variant using the approved frozen Adaptive controller, which combines a bounded RL risk signal with independent point-in-time volatility confirmation. NORMAL uses a 100% Core risk budget. DEFENSIVE can use a lower budget; Core weights are scaled by that budget and the residual is held in BIL, so defensive intensity can vary by month. The historical Adaptive comparison is reconstructed under its stated parent definition and is not presented as a live-policy record or a current-10/90 track record. The dashboard may show the current Official and Preview state, risk budget, and portfolio; training artifacts, model seeds, thresholds, and exact controller parameters remain internal. Appendix D records the PDS-specific implementation and evidence limits.

## Target and portfolio weights

| Measure                          | Meaning                                                                                | Use                                                                                                                                                     |
|----------------------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Official target                  | Portfolio weights set for the next holding period at the monthly rebalance.            | Reference for the scheduled rebalance.                                                                                                                  |
| Post-rebalance weights           | Portfolio weights immediately after the scheduled rebalance.                           | Starting weights for the next holding period.                                                                                                           |
| Pre-rebalance weights            | Portfolio weights immediately before the next rebalance after market movements.        | Used to calculate turnover and rebalancing costs.                                                                                                       |
| Current Mark                     | Provisional mark-to-market view of the incomplete holding period.                      | Monitoring only; does not change completed history or the Official Decision.                                                                            |
| Execution-day inherited exposure | The drifted portfolio carried from the prior close into the scheduled execution close. | Earns the execution-day return before the new target is applied; this can differ between Core and Adaptive even when the new close target is identical. |

### Why weight drift matters

Resetting provider weights to their strategic targets every day would define a different strategy, with different exposures, turnover, and returns. PDS therefore lets weights drift between monthly rebalances and restores the target allocation only at the next scheduled rebalance.

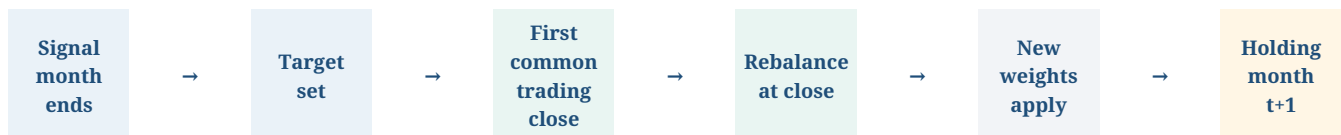
#### Key point

The provider allocation is set at the monthly rebalance and then allowed to drift; it is not reset to constant weights each day.

## TIMING

### 5. Decision and Rebalance Timing

PDS uses a monthly decision cycle. After the signal month closes, the next target is determined from information available at that cutoff. Rebalancing occurs at the first U.S. trading-day close common to both F2R and ADAA. The inherited portfolio earns the close-to-close return through that execution date; transaction cost is booked at the close; the new target starts with the next return interval. The same rule applies to Core and Adaptive. As a result, an Adaptive risk budget of 100% can still produce a different execution-day return if Adaptive entered the day with a different inherited book.



## How the monthly cycle works

- The completed signal month determines the target for the next holding period.
- The portfolio is not rebalanced at the same month-end close that produced the signal.
- The existing portfolio remains in place through the rebalance close.
- The new weights apply from the next close-to-close return interval.
- Monthly performance is labeled by the holding month in which the return is realized.
- Calendar-year performance is assigned to the year in which the holding-period returns occur.

## When dates do not line up

Strategy targets are combined only when their signal dates and relevant market-data dates are aligned. If a required strategy output is missing or not yet available, PDS does not estimate it. The last valid Official Decision remains in force until the required inputs are available and aligned.

### Why timing matters

**The signal cutoff, execution close, and first return interval of the new target are different points in time. A 100% Adaptive risk budget set at month-end means the target will match Core after execution; it does not replace the inherited Adaptive book that earns the execution-day return. This is part of the portfolio definition, not a reporting convention.**

## PERFORMANCE MEASUREMENT

## 6. Performance Measurement and Transaction Costs

All PDS performance statistics are derived from continuous daily returns. Monthly, year-to-date, annual, cumulative, drawdown, volatility, Sharpe, and Calmar figures use this daily series, with 252 trading days for annualization. Completed Performance ends at the latest completed holding month; Current MTD is provisional and shown separately. Core and Adaptive follow the same convention, but they can differ on an execution day because that day's return is earned by the pre-trade book before the new close target is applied.

$$\begin{aligned}\text{Daily wealth: } W_d &= W_{(d-1)} \times (1 + r_{d,\text{net}}) \\ \text{Monthly return: } R_{\text{month}} &= W_{\text{month-end}} / W_{\text{prior-month-end}} - 1\end{aligned}$$

### Return and risk calculations

Daily returns are the base frequency. Annualized volatility and Sharpe use 252 trading days, and drawdown is measured from the running peak of daily wealth.

### Turnover and rebalancing cost

Turnover is calculated from the portfolio weights immediately before the rebalance and the new target weights. PDS defines one-way turnover as one half of the sum of the absolute weight changes.

$$\begin{aligned}\text{One-way turnover} &= 0.5 \times \sum_i | \text{target weight}_i - \text{pre-rebalance weight}_i | \\ \text{PDS rebalancing cost} &= 2 \times \text{one-way turnover} \times 0.001\end{aligned}$$

The current cost assumption is 10 basis points per side. A full portfolio replacement (100% one-way turnover) therefore carries a 20-basis-point round-trip cost.

### Avoiding double counting

Costs already reflected in the F2R or ADAA return series are not charged again at PDS level. PDS adds only the incremental cost of moving from the drifted pre-trade book to the new PDS target. Adaptive follows the same rule and rebalances directly from its own inherited book; it does not first pay for a hypothetical Core rebalance. On an execution date, the old book earns that day's return, turnover and cost are booked at the close, and the new target applies from the next return interval.

### Consistency check

All published PDS return figures follow the same timing, turnover, cost, and daily performance rules. Any unexplained difference between views is resolved before release.

## PORTFOLIO STATUS

## 7. Official Decision and Monitoring Views

The dashboard distinguishes portfolio decisions from monitoring views. The Core Official Decision is the current Core target, while PDS Adaptive runs in parallel with its own state and risk budget. Current Mark, Preview, and Forward Shadow remain monitoring views. The public dashboard can show current Core and Adaptive state, portfolio detail, Preview, performance, and Dynamic FX information under the same timing rules as the local dashboard.

| View                       | What it shows                                                                                    | How it is used                                                                                                                         |
|----------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Core Official Decision     | The target portfolio in effect for the current holding period.                                   | Used as the PDS portfolio target. It is not a broker fill or trade confirmation.                                                       |
| Current Mark               | A provisional view of performance for the current, incomplete holding period.                    | Monitoring only; it does not change the Official Decision or completed performance history.                                            |
| Preview                    | What the next target would look like if the current month ended using the latest available data. | Monitoring only; it is not executed and no Preview performance is reported.                                                            |
| Forward Shadow             | A separately tracked alternative to the Active Core used for comparison.                         | Not traded as part of the Active Core; any performance is kept separate.                                                               |
| Public Operating Dashboard | Public reader-facing view of the current operating state.                                        | Shows current monitoring information and completed performance under the same clocks as the local dashboard.                           |
| PDS Adaptive               | Parallel Core-based variant with NORMAL or DEFENSIVE risk-control state.                         | Current state, risk budget, holdings, Preview, and performance may be shown publicly; low-level controller internals remain protected. |

### More recent does not mean official

A newer Preview or Forward Shadow result does not replace the Official Decision. It simply adds more recent monitoring information.

#### Changing portfolio status

A monitoring view becomes part of Core only through the regular monthly decision process. PDS Adaptive changes only its own parallel portfolio through the risk-control layer. A newer timestamp does not change the authority of any of these views.

## KRW INVESTOR CONTEXT

## 8. Currency Exposure and the FX Overlay

For a KRW-based investor holding U.S.-listed multi-asset ETFs, exchange-rate moves can materially affect portfolio returns and risk. PDS therefore manages currency exposure at the total-portfolio level rather than embedding a separate FX decision inside one underlying strategy.

### FX implementation and clocks

| FX view                              | Status                                          | How to interpret it                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dynamic FX operating layer           | Current monitored currency layer                | Shows the current Official and Preview monthly hedge state, Current MTD, and completed Core/Adaptive Dynamic-FX performance under the defined spot-sensitivity rule and cost schedule.                                                                                                                                                             |
| Broker-level forward/NDF realization | Not represented by the current spot-based proxy | Realized hedge P&L would require tradable forward/NDF data, actual availability, carry/basis, contract-roll rules, execution, and realized transaction costs.                                                                                                                                                                                      |
| FX clock terminology                 | Three distinct timestamps                       | Latest raw FX observation = newest source-feed timestamp; Latest admissible FX observation = newest FX point allowed by the PDS/common calendar and usable for z-score/hedge decisions; Portfolio/Preview market-through = latest date through which the relevant portfolio/Preview market inputs are complete. These dates need not be identical. |

Dynamic FX is the monthly USD/KRW implementation layer for both PDS Core and PDS Adaptive. The same FX rule, execution clock, and cost schedule apply to both; only the underlying USD portfolio path may differ. The public dashboard can show the current Official FX decision, Preview state, Current MTD, and completed FX performance. These spot-based results are an implementation proxy, not realized broker-level forward/NDF hedge P&L.

### Why the distinction matters

- Spot-FX sensitivity is not the same as realized forward/NDF hedging performance.
- Forward carry, basis, contract rolls, and changes in hedge ratios can affect actual hedging results.
- An FX series developed in a strategy research project is not automatically adopted by PDS.
- A historical FX series should be interpreted using its stated signal, admissibility, and market-through clocks rather than treated as a substitute for an executed hedge record.

## GOVERNANCE

## 9. Research Evidence, Portfolio Usefulness, and Adoption

PDS separates three questions: Does the evidence support the research claim? Does the strategy add value in the portfolio role being tested? Has it been adopted into the Official portfolio? A positive answer to one does not imply a positive answer to the others.

| Review                         | Question                                                                          | What it determines                                                                 |
|--------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Research review (Gate R)       | Does the evidence support the research claim being made?                          | Whether the research claim passes, fails, or remains inconclusive.                 |
| Portfolio-role review (Gate P) | Does the strategy add enough practical value for the portfolio role being tested? | Whether the strategy is suitable for that role, unsuitable, or still inconclusive. |
| Official adoption              | Has the strategy or allocation actually been approved for the PDS portfolio?      | Only an explicit adoption decision changes the Official portfolio.                 |

### These are separate decisions

A strategy can have sound research evidence without being useful for the role being considered. It can also look useful in a portfolio test without supporting a new research claim. Neither result changes the Official Core unless the strategy is explicitly adopted.

### When a review does not pass

If required data, timing, or evidence cannot be verified, the proposed update is not used. A Preview issue does not alter the last valid Core Official Decision. A strategy stays outside the Official Core until a later review supports adoption. Adaptive follows the same fail-closed principle: if the frozen policy, state inputs, or integrity checks fail, it produces no new decision while Core continues independently.

## MONITORING

## 10. Monitoring Views

PDS monitors the current Core Official portfolio, PDS Adaptive, Preview, Current MTD, Dynamic FX, completed performance, and portfolio history. Forward Shadow, where shown, remains a separate comparison view. Each view has a defined role and clock so monitoring information is not mistaken for a portfolio decision. After validation, the public dashboard shows the same operating information while excluding environment-specific infrastructure and protected implementation details.

### Dashboard views

| Dashboard view               | What it shows                                                                                                                 | How to interpret it                                                                                                                                |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Core                         | Current Official decision, Current MTD, provider contributions, and the primary portfolio state.                              | The Official target remains authoritative; Current MTD is provisional and separate from completed performance.                                     |
| Adaptive                     | Current state, risk budget, target relation to Core, holdings, and risk-control history.                                      | Parallel to Core. NORMAL uses 100% risk budget; DEFENSIVE can use a lower risk budget, with the residual held in BIL.                              |
| Preview                      | Core Preview and the separately evaluated Adaptive prospective state for the next holding month.                              | Target-only monitoring; not executed and excluded from completed performance.                                                                      |
| FX                           | Current Official and Preview FX state, admissible observation clock, Current MTD, hedge history, and completed performance.   | The same FX rule and cost schedule apply to Core and Adaptive USD paths; spot-based results are not broker-level NDF execution evidence.           |
| Performance                  | Core, Core + Dynamic FX, Adaptive, Adaptive + Dynamic FX, and provider/reference comparisons with stated performance periods. | Completed history remains separate from provisional Current MTD. Reconstructed PDS history and standalone provider records are labeled separately. |
| Portfolio                    | Current Core-definition, Adaptive, and monitoring target histories in fixed ETF columns, with downloadable Excel.             | Preview and Official rows are labeled explicitly. Adaptive history also shows State and Risk Budget so DEFENSIVE intensity is visible.             |
| Forward Shadow (where shown) | A separately tracked alternative to the Active Core used for comparison.                                                      | Comparison only; display or performance does not make it part of the Official Core.                                                                |

### When data are unavailable

If a strategy update, Preview input, timing reference, or Current Mark cannot be produced, PDS keeps the last valid Official information and marks the newer item as unavailable. Missing data are not estimated merely to keep the dashboard populated.

OPERATIONS AND PUBLICATION

# 11. Operating and Publication Process

PDS decisions are produced in the controlled operating environment. Publication creates a public copy of the approved dashboard rather than a separate portfolio state or decision stream. It does not create a new portfolio or trading decision; it distributes information that has already passed the operating checks.



## Main operating steps

| Step                      | Purpose                                                                                                                                                                |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Refresh and rebuild       | Update strategy inputs and rebuild the PDS dashboard. If a monitoring input is unavailable, the last valid Official Decision remains unchanged.                        |
| Review the current state  | Check the Official Decision, Current Mark, Preview, Forward Shadow, and related changes before any release is prepared.                                                |
| Prepare the public update | Generate the public dashboard, remove environment-specific infrastructure, and verify that the public version preserves the approved operating information and clocks. |
| Approve and publish       | Review the changes, confirm the published figures, and release the update to the SlackQuant site.                                                                      |

## Publication checks

1. Check the PDS public figures, labels, links, and disclosure boundaries.
2. Build the public pages and inspect the rendered result.
3. Confirm that the downloadable and linked resources match the approved release.
4. For a new system-document version, review every rendered page and fix any issue before release.
5. Publish only after the checks above pass.

### Final release approval

Automation can update data, run checks, and prepare a release, but a substantive public update still receives a human review before publication.

## PUBLIC DISCLOSURE

## 12. Public Disclosure

The public PDS dashboard is the public reader-facing counterpart of the local dashboard. It can show current Core and Adaptive state, portfolio detail, Preview, Dynamic FX monitoring, and performance. The public/private boundary is mainly technical: local paths, runtimes, caches, credentials, repository controls, debug tooling, brokerage-account information, and protected model artifacts remain private.

| Published                                                                                                                               | Internal / suppressed                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| System architecture, provider relationships, governance, and system documentation                                                       | Local filesystem paths, runtime/cache/repository locations, credentials/secrets, and debug/runner tooling                                    |
| Current Core and Adaptive decision state, detailed portfolio targets, and operating clocks                                              | Low-level Adaptive controller training artifacts, model seeds, exact thresholds/parameters, and implementation-specific internal identifiers |
| Current Mark, Preview, Forward Shadow where shown on the dashboard, and performance monitoring                                          | Brokerage-account holdings, realized account P&L, order-routing infrastructure, and trade confirmations                                      |
| Current Dynamic FX monitoring, Preview FX state, Current MTD, hedge history, and completed Dynamic-FX performance                       | Research branches and experimental artifacts that have not been promoted to the current system                                               |
| Research, Quantitative Methods, release validation, and versioned system documentation                                                  | Machine-specific metadata that is not needed to interpret the published system                                                               |
| Core, Core + Dynamic FX, Adaptive, and Adaptive + Dynamic FX performance with explicit performance periods and historical-series labels | Protected security-sensitive implementation details not required to reproduce the published operating state                                  |

### How public performance is dated

Monthly performance is labeled by the month in which the portfolio return is realized, not by the month in which the signal was formed. Annual figures follow the same holding-period convention.

### Status of public materials

Public PDS pages are research and monitoring tools, not execution interfaces or trade confirmations. Their dates and status follow the same decision, admissibility, and market-through rules used by the operating dashboard.

#### Publication rule

**A release is blocked if figures, clocks, labels, portfolio detail, or cross-system references do not agree, or if environment-specific or security-sensitive material appears in the public artifact. Publication does not change portfolio decision status.**

## RELEASE CHECKS

### 13. Release Validation

For v1.5, the current 10/90 Core construction, the separation of standalone and reconstructed histories, execution-close accounting, Adaptive state and risk-budget treatment, Dynamic FX clock terminology, dashboard terminology, and public documentation links were checked against the operating records. The table summarizes those checks; it documents the released system and is not a forecast. Appendix D contains the PDS-specific Adaptive implementation details, so this System Documentation also serves as the current technical reference for PDS.

| Check                                 | Result                                                                                                                                                                                                                                                                                    |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Core allocation                       | The current monthly top-level Core target is F2R 10% / ADAA 90%. The ratio is documented here and in the detailed dashboard, while the platform landing page emphasizes the architecture.                                                                                                 |
| Retrospective Core history            | The displayed long history applies the current 10/90 Core definition consistently across the reconstructed period. Earlier top-level mixes remain internal research references and are not spliced into the reader-facing series.                                                         |
| Standalone provider history           | F2R standalone completed performance remains its 1 May 2017 onward current-model record; ADAA standalone performance remains its 2 June 2008 onward strategy record. The longer PDS provider/reference histories are separately labeled retrospective current-definition reconstructions. |
| Rebalance timing                      | The monthly decision and execution-close dates matched the timing rules described in Section 5.                                                                                                                                                                                           |
| Transaction costs                     | Drift-based turnover and the 10 bp-per-side PDS rebalancing cost assumption matched the operating calculation; provider-level costs are not counted twice.                                                                                                                                |
| Performance measurement               | Completed Performance is derived from the daily return series with 252-trading-day annualization; Current MTD remains provisional and separate.                                                                                                                                           |
| Performance periods at release        | Core and Core + Dynamic FX share the common reader period 2 December 2005 through 31 August 2026. Adaptive fixed-policy comparison uses 3 May 2021 through 31 August 2026. Standalone provider records retain their own dates.                                                            |
| Display consistency                   | Core, Adaptive, Preview, FX, Performance, and Portfolio views matched the current terminology and operating clocks.                                                                                                                                                                       |
| Adaptive state and risk budget        | NORMAL maps to a 100% Core risk budget. DEFENSIVE may use different lower risk budgets by month; Core weights are scaled by the risk budget and the residual is allocated to BIL. State and Risk Budget are shown together in Adaptive history.                                           |
| Core isolation under Adaptive failure | PDS Core allocation and accounting remain unchanged by Adaptive. Adaptive failures are contained within the Adaptive branch and do not block or rewrite Core operation.                                                                                                                   |
| Published documentation               | The v1.5 PDF is the current System Documentation resource linked from the platform; earlier versions remain archived but are no longer the current public reference.                                                                                                                      |
| Adaptive technical detail             | PDS-specific state/action/reward design, frozen-policy binding, parent transfer, and long-history evidence limits are consolidated in Appendix D and included in the current System Documentation.                                                                                        |

## RELATED METHODS

## 14. Related Quantitative Methods

The SlackQuant Quantitative Methods library provides background on the techniques referenced in this document. The entries below are the methods most directly related to PDS.

| ID    | Method                                                      | Relevance to PDS                                                                                                                                                                                  |
|-------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| QM007 | Portfolio Backtesting and Rebalancing                       | Explains monthly rebalancing, weight drift, and next-period return timing.                                                                                                                        |
| QM008 | Drawdown, Underwater Duration, and Recovery                 | Covers drawdown depth, duration, recovery, and running peaks.                                                                                                                                     |
| QM009 | Turnover, Transaction Costs, and Net Performance            | Explains turnover, transaction costs, and net portfolio returns.                                                                                                                                  |
| QM011 | Strategy Similarity and Decision Diversification            | Provides the basis for comparing strategy decisions rather than return correlation alone.                                                                                                         |
| QM013 | Benchmark-Relative Portfolio Evaluation                     | Covers active return, tracking error, information ratio, relative wealth, and relative drawdown.                                                                                                  |
| QM014 | Information Timing, Data Vintages, and Real-Time Macro Data | Explains release timing, data vintages, revisions, and the information available at each decision date.                                                                                           |
| QM028 | Reinforcement Learning for Adaptive Portfolio Risk Control  | General background on RL-based risk control. Appendix D records the PDS-specific state/action/reward design, frozen-policy deployment, transfer boundary, execution linkage, and evidence limits. |
| QM029 | Dynamic Currency Hedging and FX Overlay Accounting          | General background on FX overlays. The PDS-specific raw/admissible observation rules, market-through clock, monthly hedge decision, and Core/Adaptive treatment are defined here.                 |

**Methods library:** [research.slackquant.com/methods/](https://research.slackquant.com/methods/)

### How these references are used

The Methods library explains the general techniques; this document records the PDS implementation. Read QM028 together with Section 4 and Appendix D for Adaptive risk control, and QM029 together with Sections 8 and 13 for the Dynamic FX operating contract.

## LIMITATIONS

## 15. Limitations and Revision Policy

### Limitations

- Historical results, including observations from live operation, do not guarantee future returns.
- The current F2R/ADAA allocation is the present PDS Core policy. It is a system design choice, not a claim that the ratio is universally or permanently optimal.
- F2R and ADAA can still share market exposures even though their decision processes differ. Their combination should therefore be evaluated from the observed portfolio results rather than assumed to be diversified by design.
- Transaction costs are modeled assumptions. Actual trading costs can differ.
- Dynamic FX uses spot-FX sensitivity and an assumed hedge-renewal cost schedule. It should not be read as realized broker-level forward/NDF hedging performance.
- Preview and Forward Shadow are monitoring tools and do not change the Official portfolio unless they are separately adopted through the regular decision process.
- The Adaptive fixed-policy historical comparison comes from exploratory research and should not be read as independent confirmatory evidence. Operational Adaptive decisions and realized accounting are tracked separately.
- The frozen Adaptive controller was trained under the prior 25% F2R / 75% ADAA parent definition and is now applied to the 10% / 90% Core without automatic retraining. Historical results from the prior parent are therefore not presented as a current-Core track record.
- This document describes the PDS portfolio framework. Broker fills, custody records, account holdings, and trade confirmations are outside its scope.

### When a new version is needed

| Change                                                                                                         | Update needed                                             |
|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Presentation-only edit                                                                                         | Normally no new version unless the meaning changes.       |
| Routine data update or new completed month                                                                     | No new version; the historical record is simply extended. |
| Core strategy membership or target allocation                                                                  | New document version.                                     |
| Rebalance timing, turnover, or transaction-cost treatment                                                      | New document version.                                     |
| Definition or role of Official Decision, Preview, or Forward Shadow                                            | New document version.                                     |
| Public disclosure policy                                                                                       | New document version.                                     |
| Dynamic FX role, implementation method, or accounting treatment                                                | New document version.                                     |
| PDS Adaptive role, policy binding, evidence boundary, execution semantics, or performance-accounting treatment | New document version.                                     |

### Versioning rule

A new version is issued when portfolio construction, timing, cost treatment, decision status, or disclosure policy changes materially. Routine monthly data updates do not require a new version.

## Appendix A. Decision and Publication Status

| View                   | Current?                              | Can change Core target?                                                             | Performance?                                                                            | Published?                                                                                                    |
|------------------------|---------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Core Official Decision | Yes                                   | Yes - through the regular monthly decision process                                  | Realized under the execution-close accounting rule                                      | Yes - current target and operating clocks are shown in the public dashboard                                   |
| Current Mark           | Yes - incomplete period               | No                                                                                  | Provisional current-period view                                                         | Yes - monitoring status is explicit                                                                           |
| Preview                | Yes - incomplete period               | No                                                                                  | No executed Preview performance                                                         | Yes - shown as target-only and non-authoritative.                                                             |
| Forward Shadow         | Yes - monitoring                      | No                                                                                  | Separate comparison performance may be tracked                                          | Yes where shown on the public dashboard; display alone does not change its status                             |
| Public Dashboard       | Current published snapshot            | No - publication does not change portfolio decisions                                | Shows current monitoring information and completed performance                          | Yes                                                                                                           |
| Dynamic FX             | Yes - separate monthly currency layer | No - does not change the underlying Core target                                     | Current MTD and completed overlay performance                                           | Yes - current Official/Preview state and completed performance; broker-level NDF execution is not implied     |
| PDS Adaptive           | Yes - parallel operating variant      | Changes only the Adaptive target through its risk-control layer; never changes Core | Fixed-policy historical comparison plus separate post-deployment operational accounting | Yes - current state, risk budget, and portfolio can be shown; low-level controller internals remain protected |

### Reading the table

- The Core Official Decision changes the Core target through the regular monthly process. PDS Adaptive changes only its own parallel target through the risk-control layer. Preview, Forward Shadow, and Current Mark remain monitoring views. Publication does not turn a monitoring view into an Official Decision.

## Appendix B. PDS Resources

| Resource                 | Purpose                                                                                                                                                                               | Location                                                                                                            |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| PDS System page          | Overview of the system, PDS Core / PDS Adaptive architecture, current Core providers, and related public resources.                                                                   | <a href="https://research.slackquant.com/systems/pds/">research.slackquant.com/systems/pds/</a>                     |
| PDS Dashboard            | Public operating view for Core, Adaptive, Preview, FX, Performance, and Portfolio.                                                                                                    | <a href="https://research.slackquant.com/systems/pds/dashboard/">research.slackquant.com/systems/pds/dashboard/</a> |
| PDS System Documentation | Authoritative explanation of Core/Adaptive construction, timing, accounting, monitoring, governance, disclosure, and the consolidated PDS-specific Adaptive technical implementation. | Linked from the PDS System page                                                                                     |
| Quantitative Methods     | Background on the quantitative methods referenced in this document.                                                                                                                   | <a href="https://research.slackquant.com/methods/">research.slackquant.com/methods/</a>                             |
| Research                 | Separate empirical papers and research records related to the strategies and methods.                                                                                                 | <a href="https://research.slackquant.com/research/">research.slackquant.com/research/</a>                           |

Start with the System page for an overview. The Dashboard shows the public operating view for current Core, Adaptive, Preview, FX, Performance, and Portfolio information. This document and the Methods library provide the supporting system and methodology detail.

## Appendix C. PDS v1.5 Summary

| Item                        | PDS v1.5                                                                                                                                                                                                                                                              |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| System                      | Portfolio Decision System (PDS)                                                                                                                                                                                                                                       |
| System type                 | Portfolio Decision & Operating System                                                                                                                                                                                                                                 |
| Current Core providers      | F2R and ADAA                                                                                                                                                                                                                                                          |
| PDS Core                    | Base portfolio with a monthly top-level target of F2R 10% + ADAA 90%. The platform landing page emphasizes the architecture rather than the ratio.                                                                                                                    |
| Rebalance frequency         | Monthly                                                                                                                                                                                                                                                               |
| Between rebalances          | Provider and asset weights drift with market returns; there is no daily reset to strategic provider targets.                                                                                                                                                          |
| Rebalance timing            | First common U.S. trading-day close after the signal month. The inherited pre-trade book earns through that close; the new target starts from the next close-to-close interval.                                                                                       |
| Turnover measure            | One-way turnover is half the sum of the absolute changes between pre-rebalance and target weights.                                                                                                                                                                    |
| Transaction-cost assumption | 10 basis points per side.                                                                                                                                                                                                                                             |
| Performance calculation     | One continuous daily return series; monthly and annual results are derived from it.                                                                                                                                                                                   |
| Preview                     | Monitoring only; not executed and no Preview performance is reported.                                                                                                                                                                                                 |
| History separation          | Standalone provider performance remains separate from the longer retrospective PDS current-definition reconstruction. The reconstructed series is for research/comparison and is not presented as historical live-policy operation.                                   |
| Public disclosure           | Public and local dashboards present the same reader-facing operating information. Environment and security infrastructure, along with protected implementation details, remain private.                                                                               |
| Dynamic FX                  | Third operating layer: monthly USD/KRW overlay applied under the same rule, execution clock, and cost schedule to Core and Adaptive USD paths. Current Official/Preview state and completed performance may be displayed.                                             |
| FX evidence boundary        | Spot-based operating sensitivity with assumed renewal cost is not broker-level forward/NDF realized P&L. Raw FX observation, admissible FX observation, and portfolio/Preview market-through are distinct clocks.                                                     |
| PDS Adaptive                | Parallel Core-based variant using Reinforcement-Learning Adaptive Risk Control; RL provides bounded risk control rather than standalone alpha or security selection.                                                                                                  |
| Adaptive operating states   | NORMAL or DEFENSIVE. NORMAL uses a 100% Core risk budget; DEFENSIVE can use a lower risk budget that varies by month, with residual allocation to BIL. Current state and risk budget may be shown publicly; controller recipe and training artifacts remain internal. |

| Item                       | PDS v1.5                                                                                                                                                                                                                                                                                                 |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Adaptive evidence boundary | Historical Adaptive comparison is a fixed-policy reconstruction over its stated evaluation period. Current Official/Preview decisions and post-deployment operational accounting remain separate. It is not presented as a retrospective live-policy record and is not spliced into operational history. |
| Adaptive failure behavior  | Fails closed within the Adaptive branch; PDS Core remains independent and continues to refresh.                                                                                                                                                                                                          |

**About this version**

Version 1.5 describes the PDS framework in effect on 27 September 2026. The operating architecture and current Core policy are unchanged from v1.4. This release adds the durable PDS Adaptive technical implementation detail to Appendix D: state/action/reward design, frozen-policy binding, parent transfer, and long-history evidence limits. It does not retrain the RL policy, alter Core/Adaptive/FX decisions, or promote research-only long-history RL diagnostics to operational evidence.

## Appendix D. PDS Adaptive Technical Implementation

This appendix records the PDS-specific implementation detail needed to understand Adaptive. Sections 1-15 remain the operating contract; Appendix D is explanatory and does not change portfolio authority, controller parameters, or evidence class.

### D.1 Why the controller is hybrid

RPS research began with a broader controller that could alter both the F2R/ADAA mix and total portfolio risk. Subsequent tests asked whether any benefit required sequential RL or could be explained by simpler risk control. Because simple de-risking and point-in-time volatility control remained credible comparators, the implemented design was narrowed to a bounded risk-only RL signal inside a hybrid controller. Current evidence does not establish that RL is generally superior to volatility targeting.

| Research step                     | Question                                                               | Design consequence                                                           |
|-----------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Broad controller                  | Should RL control provider mix and total risk?                         | Useful research baseline, but attribution remained too broad for production. |
| Sequentiality ablation            | Does future credit assignment add information beyond a myopic version? | Sequential signal retained for later research and the frozen controller.     |
| Simple risk-control falsification | Can simple de-risking explain the apparent gain?                       | Pure-RL deployment was not justified.                                        |
| Risk-channel benchmark            | Does risk-only PPO dominate strong same-channel controls?              | Evidence was mixed; volatility control remained a strong comparator.         |
| Deterministic ensemble            | Can seed dependence be reduced at the production interface?            | A fixed multi-policy deterministic ensemble was adopted.                     |

### D.2 State, action, and reward

At each completed signal-month cutoff, the frozen controller receives an ordered point-in-time state vector. Standardization uses means and standard deviations stored in the frozen model bundle; production inference does not refit them on later history.

| State family                 | Information represented                                               | Purpose                                                            |
|------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------|
| Market trend and breadth     | Multi-horizon equity and cross-asset trend, breadth, dispersion       | Describe directional and cross-sectional market state.             |
| Realized risk and dependence | Equity/bond volatility, stock-bond and broad cross-asset dependence   | Measure current risk concentration and diversification conditions. |
| Provider behavior            | F2R and ADAA trailing return, volatility, drawdown, relative behavior | Describe how the underlying providers are behaving.                |
| Provider disagreement        | Target-book distance, overlap, defensive-asset weights                | Measure convergence or disagreement between provider decisions.    |
| Policy memory                | Previous target weights across F2R, ADAA, and BIL                     | Carry prior action information without execution-day lookahead.    |

Execution-close pre-trade drift, later revisions, future-known information, and holding-period outcomes are excluded from the month-end state. Previous target weights carry policy memory without execution-day lookahead.

**Bounded action**

Let  $\rho$  denote the monthly Core risk budget, bounded between 0 and 1. Adaptive target =  $\rho \times \text{Core target} + (1 - \rho) \times \text{BIL}$ . At  $\rho = 1$ , Adaptive equals Core. The controller does not re-rank securities or alter the provider mix.

The deployed controller combines the deterministic output of the frozen PPO ensemble with an independent point-in-time volatility-control budget. Both signals must indicate caution before the hybrid controller enters a defensive state. Exact ensemble statistics, trigger thresholds, the defensive floor, feature lookback grid, and low-level tuning constants remain protected implementation details.

**Actual PPO training reward**

The selected risk-only PPO was trained on transaction-cost-adjusted net log wealth. It does not add separate volatility, drawdown, or turnover penalties to the reward. The risk-control character comes mainly from the bounded action channel, residual BIL allocation, and hybrid confirmation rule; risk and return metrics are used to evaluate candidates after training rather than inserted as extra PPO reward penalties.

**D.3 Frozen-policy training, certification, and inference**

Training follows monthly chronology. A transition enters the training sample only when its outcome is known before the out-of-sample cutoff, and state normalization is fitted on the same pre-OOS history. The selected model family is a compact feed-forward actor-critic with a bounded Beta action distribution; deterministic deployment uses the distribution mean. Hidden widths, optimization counts, learning-rate constants, discount settings, seed identities, and similar tuning details are not public.

| Certification layer         | What is checked                                                                                                                     |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Model identity              | Architecture, training metadata, feature order, normalization arrays, and training support.                                         |
| Artifact integrity          | Model files and manifest are hash-bound; missing or changed artifacts fail closed.                                                  |
| Numerical / decision parity | Regenerated continuous outputs stay within tolerance and discrete NORMAL/DEFENSIVE decisions match the approved research reference. |
| Portfolio parity            | Target, daily return, wealth, cost, and ending-weight paths reconcile under tight tolerances.                                       |
| Reload parity               | Saved and reloaded frozen models reproduce deterministic inference.                                                                 |

The operational module does not retrain the policy. If the frozen bundle, state contract, or required inputs fail integrity checks, the Adaptive branch fails closed and PDS Core remains unchanged.

Monthly inference follows a fixed sequence: build the admissible month-end state, apply frozen standardization, carry the previous target as policy memory, run deterministic inference across the frozen policies, aggregate the RL risk-budget signal, compute the independent PIT volatility budget, apply the frozen hybrid rule, scale the current Core target and assign the residual to BIL, then execute at the same first common U.S. trading-day close used by Core. Dynamic FX, when measured, is applied afterward to the realized USD path under the common FX rule.

**Current parent transfer**

The frozen controller was trained under the prior 25% F2R / 75% ADAA parent definition. Applying it to the current 10% / 90% Core is an explicit transfer decision, not an automatic retraining run. Historical prior-parent results are therefore not presented as a current-Core Adaptive track record.

#### D.4 Evidence boundary after long-history review

- The certified frozen-policy historical comparison still begins on 3 May 2021 and runs through 31 August 2026 for the current release. The longer Core reconstruction does not move that Adaptive boundary backward.
- Backward replay of the current frozen controller over earlier reconstructed history is explicitly diagnostic stress replay. Because the current policy was learned later, the replay is not genuine historical out-of-sample evidence.
- The fixed-36 cross-vintage R&D trains annual historical policies using exactly the immediately preceding 36 completed transitions and evaluates the distribution across vintages rather than selecting a best window. The resulting evidence shows training-regime sensitivity and does not establish general RL superiority over strong simple risk controls.
- Long-history RL diagnostics, action-attenuation and action-floor screens, and cross-vintage experiments remain research-only unless a separately governed adoption decision changes production authority. None of these research outputs is promoted automatically in v1.5.

Consistent with these limits, Adaptive remains a bounded parallel risk-control branch; it does not replace or rewrite Core. The operating decision record remains separate from selection-aware historical research evidence.

| Publicly documented                                                     | Implementation-protected                                                              |
|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Adaptive role and bounded risk-budget semantics                         | Exact ensemble statistic, trigger thresholds, and defensive floor                     |
| State families and point-in-time information boundary                   | Exact feature lookback grid, feature ordering details, and low-level tuning constants |
| Net log-wealth reward form and transaction-cost treatment               | Model seed identities, private model artifacts, and training checkpoints              |
| Frozen-policy concept, hash binding, and parity checks                  | Private runtime paths, credentials, caches, and debug infrastructure                  |
| Historical vs operational evidence classes and parent-transfer boundary | Implementation-specific identifiers not required for interpretation                   |

QM028 remains the general method reference for RL-based portfolio risk control. This appendix records only the PDS-specific implementation choices and evidence limits. QM029 remains the general method reference for Dynamic FX; the PDS-specific FX operating contract is defined in Sections 8 and 13.

### SlackQuant Research

System documentation only. Not investment advice.